THE DESIGN QUALITY CODEX

A universal, taste-neutral standard for building premium digital interfaces.

Feed this document to any human designer or AI coding agent at the start of any project — web app, mobile app, landing page, dashboard, e-commerce, tool — and hold every screen to it. It encodes interaction research (Nielsen Norman Group, Baymard Institute, ACM HCI literature), normative standards (WCAG 2.2), platform guidance (Material 3, Apple HIG, Fluent 2), and hard-won production lessons into one enforceable rulebook.

> **Taste neutrality.** This codex does not prescribe a style. Minimalism, maximalism, brutalism, glassmorphism, editorial, corporate, playful — every aesthetic can be premium and every aesthetic can be slop. Premium is not a look; it is a *property of execution*: every pixel decided, every state designed, every interaction acknowledged, every number justified. Layout patterns (split heroes, card grids, sidebars) are never banned here — only *undecided, default-driven execution* of them is.

> **Evidence grading used throughout: [LAW]** = peer-reviewed research or W3C normative standard · **[STRONG]** = large-sample industry research (NN/g, Baymard) · **[HEURISTIC]** = widely-practiced professional convention · **[FLAGGED]** = popular claim with unverifiable numbers; treat with suspicion.

PART 0 — WHAT SLOP ACTUALLY IS (and what premium actually is)

0.1 The definition

Slop is what undecided pixels look like. It is not a particular layout, font, or color. It is the visible residue of decisions nobody made: the default font because no one chose one, the third identical section because no one asked if it earned its place, the missing loading state because no one imagined slow networks, the `ease` timing function because no one felt the motion. AI systems produce slop at scale because they produce *plausible defaults* at scale — and a plausible default is precisely a decision nobody made.

0.2 The symmetry test

Any pattern can be premium or slop. A split hero (text left, image right) executed with a deliberate type scale, an art-directed image, asymmetric tension, and a reason for being split — premium. The same split hero with a stock illustration, default spacing, and centered gradient blob — slop. **The pattern is innocent; the execution is guilty.** Therefore this codex regulates *execution qualities* (intent, consistency, completeness, feedback, craft) and leaves *stylistic choices* (layouts, palettes, moods) free.

0.3 The five universal execution failures (taste-independent slop signals)

These are detectable in any style, and each has a universal cure:

#	Failure	How to detect it	Universal cure

1	**Undecided defaults**	Framework default font/radius/shadow/palette present with no evidence of choice	Every foundational token (type, color, spacing, radius, shadow, easing) is explicitly chosen and recorded, even if the choice equals the default — chosen-default ≠ unexamined-default
2	**Uniform weighting**	Every section/card/element has equal size, equal emphasis, equal rhythm — nothing is more important than anything else	Establish hierarchy: one dominant element per screen; sizes, weights, and spacing that *rank* content; if everything shouts, nothing speaks
3	**Incomplete states**	Only the ideal state exists; loading/empty/error/edge states are absent or accidental	The full state inventory (Part 9) designed for every dynamic surface before a screen is called done
4	**Silent or incoherent feedback**	Actions with no acknowledgment; animations with mixed easing/duration DNA; hover on some buttons but not others	Feedback contract (Part 7) on 100% of interactive elements; one motion system (Part 8) governing all movement
5	**Copy that decorates instead of works**	Generic adjectives ("seamless", "powerful", "blazing fast"), headings that repeat the section, placeholder-tone microcopy	Copy is interface: specific claims with real numbers, verbs over nouns, one idea per sentence, microcopy at every anxiety point (Part 14)

0.4 What premium is

Premium = **intent × consistency × completeness × feedback × restraint**, perceivable within 50 milliseconds and confirmed over 50 interactions.

- **Intent:** every element can answer "why are you here, why do you look like this?"
- **Consistency:** one type system, one spacing rhythm, one motion DNA, one color logic — self-similar at every zoom level.
- **Completeness:** the product is designed for its worst day (slow network, empty account, 300-character titles), not just its screenshot day.
- **Feedback:** the interface acknowledges the user's existence instantly, always, proportionally.
- **Restraint:** effects, colors, and motion are budgets to be spent on meaning, not confetti. The most premium products spend less, more precisely.

0.5 The one-sentence brief every project must produce

Before any screen is drawn, write one sentence: *"This product makes [audience] feel [emotional quality] while doing [job], and the one thing they should remember is [signature]."* Every subsequent decision is auditable against this sentence. An interface that cannot state its sentence will look like it cannot — that is where slop begins.

PART 1 — OPERATING PRINCIPLES (process rules that prevent slop before pixels exist)

1. **Decide the foundations first, in writing.** Type scale, palette with roles, spacing unit, radius scale, shadow scale, easing curves, duration scale. Ten minutes of token decisions prevents a thousand micro-inconsistencies. A named system gets respected; anonymous values drift.

2. **Give every color, easing, and effect a role, not just a value.** "Accent = confirmation and primary action, nowhere else" is a system. "Accent = #6366F1" is a paint bucket. Roles make a thousand later decisions automatic and make violations detectable.
3. **One dominant element per screen.** Decide what the user should see first, second, third — then verify with the squint test (blur your eyes: does the intended hierarchy survive?).
4. **Design the worst day.** Slow connection, empty account, expired session, longest possible content, smallest supported screen, keyboard-only user. Premium is measured on the worst day; anything looks good on the screenshot day.
5. **A signature moment per key surface [HEURISTIC].** One interaction, transition, or detail the user could describe to a friend. It can be tiny (a perfect toggle) or theatrical (a ceremony on a destructive action) — style-appropriate, but present. Products with zero memorable moments are forgotten by definition.
6. **Conventions are load-bearing.** Users import expectations from every other product they use (Jakob's Law, Part 5). Spend novelty on content and signature moments; keep mechanics (navigation, links, back button, form behavior) boringly conventional unless there is a defended reason.
7. **Copy is designed with the layout, not poured in after.** Lorem ipsum hides hierarchy failures. Real words, real lengths, real numbers from day one.
8. **Verify hostile, twice** (full ritual in Part 16). Quality is not the first pass; it is the second pass after the first pass's failures are fixed.

PART 2 — TYPOGRAPHY (the 80% of interface design)

2.1 Family selection

- **Two families maximum** as a default discipline (one workhorse UI/body, one optional display voice). A monospace for data/code/labels counts as a voice — deliberate, not incidental. More families are possible in editorial styles but each must have a stated role.
- **Choosing the default is allowed; not choosing is not.** System stacks and popular fonts are legitimate when *chosen* for a reason (performance, neutrality, platform-nativeness) and configured (sizes, weights, spacing) rather than left at framework defaults.
- Check the font at the sizes you will actually use: some faces collapse at 12px labels or fall apart at 96px display. Test both extremes before committing.

2.2 Scale

- **Build a modular scale with real contrast.** Timid scales (16/18/20/24) produce wallpaper. Confident hierarchies jump: small labels (10–12px) → body (16–18px) → subheads (22–28px) → display (48–100+px via `clamp()`), tuned to the product's genre (dashboards compress the top, marketing stretches it).
- **Fluid display sizes:** `font-size: clamp(min, preferred-vw, max)` so headlines breathe with the viewport instead of snapping at breakpoints.
- **Line length: 45–75 characters** for body text **[STRONG]**; enforce with `max-width` in `ch` units. Full-width paragraphs on desktop are a top-3 amateur tell in any style.
- **Line height inversely proportional to size:** ~1.5–1.7 body, ~1.2 subheads, 0.95–1.15 display. Large type with body-text line-height reads as unset.
- **Letter-spacing inversely proportional to size:** slightly negative for display (-0.01 to -0.04em), neutral for body, generously positive (+0.08 to +0.5em) for small uppercase labels. This single pairing — tight display

+ tracked micro-labels — is the cheapest craft signal that consistently works across styles.

2.3 Typographic craft details

- **16px minimum body on mobile**; inputs ≥16px or iOS zooms on focus and stays zoomed.
- **Tabular numerals** (`font-variant-numeric: tabular-nums`) for tables, counters, timers — proportional digits jiggle when values change.
- Real punctuation: curly quotes, en/em dashes, real ellipsis, real multiplication sign in "4×4". Middle-truncate identifiers and filenames (`abc…xyz`), end-truncate prose.
- Orphan control on headings (no single word alone on the last line — use `text-wrap: balance` where supported).
- Font loading: `font-display: swap` plus a metric-compatible fallback to keep CLS ≈ 0 (Part 15).
- Hierarchy is multi-channel: size, weight, color, spacing, and case are five separate levers — premium typography rarely pulls only the size lever.

PART 3 — COLOR

3.1 Building a palette that cannot go wrong

- **Choose one neutral temperature and stay inside it.** Warm neutrals (cream, bone, warm grey) or cool neutrals (porcelain, blue-grey) — mixing temperatures across surfaces is the most common invisible reason a UI "feels off".
- **Roles before values.** A complete palette assigns: surface (background layers, usually 2–4 steps), ink (text hierarchy, 2–3 steps), structure (borders/hairlines), primary action, and semantic set (success / warning / danger / info). Then — and only then — pick hex values.
- **The accent budget [HEURISTIC — 60/30/10 rule of thumb]:** dominant neutral field, supporting structure, and a small accent budget spent on *meaning* (actions, states, verdicts) rather than decoration. The scarcer the accent, the more premium it reads — true in every style including maximalism, where the "accent" may be a whole secondary palette but is still governed.
- **Semantic colors are contracts.** If red means destructive/error, red may not also decorate a marketing badge. Users learn color meanings within minutes; breaking the contract silently is a trust leak.

3.2 Normative contrast [LAW — WCAG 2.2 1.4.3 / 1.4.11]

- Body text: **4.5:1** minimum against its background.
- Large text (≥24px, or ≥18.5px bold): **3:1**.
- UI component boundaries and meaningful graphics: **3:1**.
- Check *real* combinations (placeholder text on input background, text over images/gradients at their lightest point), not just the palette swatches.

3.3 Dark themes and gradients (execution rules, not bans)

- Dark UIs: avoid pure #000 fields with pure #FFF text (halation); use near-blacks with slightly desaturated text; elevation is communicated by *lightening surfaces*, not by shadows.
- Gradients: legitimate as a deliberate identity choice; slop when they are the *substitute* for a decision (the reflexive purple-to-blue wash behind a hero). Rules of execution: control both endpoints' contrast against

overlaid text, avoid banding (subtle noise/dither on large gradients), and give the gradient a role like any other color.

- Low-contrast, muted palettes are premium *only* with compensating structure (hairlines, clear type hierarchy). Softness without structure reads as emptiness — calm and empty are different properties.

3.4 Color accessibility beyond contrast

- **Never encode meaning in color alone** [LAW — WCAG 1.4.1]: pair color with a word, icon, or pattern ("● Failed" not just red).
- Test with common color-vision deficiencies (protanopia/deuteranopia simulators); red/green status pairs must survive.

PART 4 — SPACING, LAYOUT & STRUCTURE

4.1 Spacing as a system

- **One base unit (4 or 8px) and a geometric-ish scale** (4, 8, 12, 16, 24, 32, 48, 64, 96, 128…). Arbitrary values (13px, 27px) are the fingerprint of unsystematic work.
- **Spacing communicates grouping [LAW — Gestalt proximity]:** related things closer, unrelated things farther — and the ratio matters: within-group spacing should be clearly smaller (≈half or less) than between-group spacing. Group with space before boxes, boxes before color fills.
- **Section rhythm:** long pages need a consistent macro-rhythm (e.g., a fixed large gap between major sections) — this single decision makes pages feel composed instead of stacked.
- **Whitespace is a frame, not a substitute.** Generous space reads as luxury only when it frames crafted content; space around thin content reads as unfinished. Density and calm are independent axes: a fine-print certificate is dense AND calm; a sparse page can be chaotic.

4.2 Layout discipline

- Content max-widths always: prose ~60–75ch, forms ~400–560px, marketing containers ~1050–1320px. Full-bleed is a deliberate moment, not a default.
- **Alignment is binary: things either align or they don't.** Pick grid lines and audit every screen edge-to-edge; 2px drift between sibling elements is subliminally detectable and reads as cheap.
- Establish and repeat a radius scale (e.g., 4/8/16/full) and a shadow scale (2–3 elevations); mixed radii and one-off shadows are uniform-weighting failures in disguise.
- Every viewport-height of a scrolling page should introduce a new *kind* of element or answer a new question — momentum design. Three identical section skeletons in a row is a hierarchy failure regardless of how each looks alone.
- Responsive strategy is *reflow with intent*, not shrink-to-fit: decide content priority per breakpoint; the mobile order is an editorial decision (Part 10).

PART 5 — THE RESEARCH CANON (laws every builder must know, with provenance)

5.1 Response-time thresholds [STRONG — Nielsen 1993, still canonical]

- **0.1 s** — perceived as instantaneous. Budget for acknowledging any direct manipulation (press, hover, drag).
- **1 s** — limit for keeping the user's flow of thought. Delay is noticed but tolerated without feedback beyond the result.
- **10 s** — limit for keeping attention at all. Anything near it needs progress indication and an escape hatch.

Derived rules: <1s show nothing extra; 1–10s show skeleton/indeterminate indicator; >10s (or known-size work) show determinate progress with steps **[STRONG — NN/g progress indicators]**.

5.2 Jakob's Law [STRONG — NN/g]

Users spend most of their time in *other* products; they arrive with wholesale expectations about how navigation, links, carts, forms, and back buttons behave. Innovation budget goes to content and signature moments; convention budget covers mechanics. Violating a convention requires a defended reason and visible affordance re-teaching.

5.3 Fitts's Law [LAW — Fitts 1954]

Target-acquisition time grows with distance and shrinks with target size. Applications: primary actions large and near the locus of attention; screen edges/corners are infinitely deep targets on desktop; on touch, the thumb's home area (bottom-center) is the premium real estate (Part 10.1).

5.4 Hick's Law [LAW — Hick 1952]

Decision time rises with the number and complexity of choices. Applications: one primary CTA per view, progressive disclosure, opinionated defaults for the 80% case. Caveat: applies to *deciding*, not to scanning familiar lists — do not use it to justify hiding navigation (Jakob wins).

5.5 Aesthetic–Usability Effect [LAW — Kurosu & Kashimura 1995]

Users perceive more attractive interfaces as more usable and forgive their flaws. Craft literally buys tolerance. Danger: it blinds *you* during testing — beautiful prototypes hide usability failures; that is why hostile verification (Part 16) is mandatory.

5.6 First impressions in ~50 ms [LAW — Lindgaard et al. 2006]

Visual-appeal judgments form before reading and anchor subsequent judgments (confirmation bias). The first viewport has one job: pass the 50ms test with clear quality signals and one dominant element.

5.7 Miller's 7±2 — correctly applied [LAW — Miller 1956]

The finding is about *working memory chunks*, not menu lengths. Correct use: chunk long strings (card numbers, IDs, phone numbers) and group related information. Incorrect use: capping visible list items at 7 — scanning is not memorization.

5.8 Doherty Threshold [HEURISTIC — Doherty & Thadani 1982]

Engagement rises sharply when system response stays under ~400ms in interactive loops. Practical round-trip target, not physics.

5.9 Peak–End Rule [LAW — Kahneman & colleagues]

Experiences are remembered by their most intense moment and their ending, not their average. Engineer the peak (signature moment) and the endings: success screens, confirmations, empty states after cleanup, even the 404. Cheap surfaces to invest in, disproportionate memory returns.

5.10 Goal-Gradient Effect [LAW — Hull 1932; Kivetz et al. 2006]

Motivation increases as people approach a goal. Applications: progress indicators that show earned progress (onboarding checklists starting at 1 of 5 complete, not 0), multi-step flows that show proximity to finish.

5.11 Zeigarnik Effect & recognition over recall [LAW]

Incomplete tasks are remembered better (Zeigarnik) — surface unfinished states ("Draft — 2 fields left"). Recognition beats recall (NN/g heuristic #6) — show options, recently used items, and previews instead of demanding memory.

5.12 NN/g's 10 usability heuristics (the shared review vocabulary)

1. Visibility of system status · 2. Match between system and real world · 3. User control and freedom (undo) · 4. Consistency and standards · 5. Error prevention · 6. Recognition rather than recall · 7. Flexibility and efficiency of use · 8. Aesthetic and minimalist design · 9. Help users recognize, diagnose, recover from errors · 10. Help and documentation.

Use them to *name* problems in reviews ("this violates #1: no status during upload") — naming forces precision and removes taste arguments.

5.13 How to research (source hierarchy for any design question)

- **Normative / decide-on-this-alone:** W3C WCAG, platform guidelines (Material 3, Apple HIG, Fluent 2) for their platforms.
- **Strong evidence:** Baymard Institute (large-N usability, esp. e-commerce/forms/checkout), NN/g articles, peer-reviewed HCI (ACM CHI, TOCHI).
- **Inspiration only — steal moods, never conclusions:** Dribbble, Awwwards, Mobbin, Behance, screenshot galleries. Presence in a gallery proves someone shipped it, not that it works.
- **Numbers hygiene:** viral design statistics ("users read only 20%", "progress bars increase conversion 86%") are usually **[FLAGGED]** — unverifiable or misquoted. Cite the graded sources above or label the claim a heuristic.

PART 6 — SATISFACTION INTERACTIONS (micro-interactions that feel expensive)

6.1 Anatomy [Dan Saffer's model]

Every micro-interaction has four designable parts — slop interactions have only the first two:

1. **Trigger** — user action (tap, hover, scroll, drag) or system event (upload complete, message arrived).

2. **Rules** — what changes, what is allowed, what state results.
3. **Feedback** — how the user learns what happened: visual, haptic, audio (Part 7).
4. **Loops & modes** — behavior over time and repetition: does the 50th occurrence shorten or get out of the way? Should it change when a mode is active?

Design all four, explicitly, for every interactive element that matters.

6.2 The properties that make an interaction *satisfying* (style-independent)

- **Visible causality.** The thing touched is the thing that responds, and the response originates at the point of contact (ripple from the tap point, menu grows from its trigger, deleted row collapses in place). Effects that appear elsewhere break the illusion of direct manipulation.
- **Physicality.** Ease-out arrivals, slight mass and momentum, optional 1–2% overshoot on playful brands, none on sober brands. Linear motion for UI movement is the loudest "nobody felt this" tell in any aesthetic.
- **Proportionality.** Feedback magnitude matches action significance: a toggle clicks subtly; a destructive confirmation may earn a ceremony; a routine save earns almost nothing. Over-celebrating trivial actions is the interaction equivalent of exclamation-point copy.
- **Interruptibility.** Animations never block input. The user can act mid-animation and the system skips to the consequence. An animation you must wait for is a bug with production values.
- **Anticipation and payoff.** The best interactions have a micro-story: press → tension → release → consequence. Even a 150ms button press has these beats (down-state, release, result).
- **Earned, not ambient.** Motion as a *response to intent* (hover, press, arrival in view), not as permanent background activity. Perpetual floating/pulsing elements tax attention and read as template demos.
- **Diminishing choreography.** First-run moments may be theatrical; the hundredth repetition must cost zero patience. Frequency-sensitive design: rare + significant = more theater; frequent + trivial = less.

6.3 A universal pattern library (adapt the skin to any style)

- **State ceremonies** for rare, weighty actions: publish, delete-forever, sign, pay. A deliberate 400–1000ms sequence with distinct beats. Gravity must match consequence — never ceremonial trivia.
- **Draw-on/verify moments:** SVG stroke self-drawing, checkmark drawing on success — one per surface, not per item.
- **Hover elevation (desktop):** small translateY + shadow/边 change + one accent detail, 200–400ms ease-out. Never scale blocks containing text (mid-transition blur).
- **Press acknowledgment:** scale 0.97–0.98 or darken, <100ms, pure CSS `:active`.
- **List/row response:** hover tint + slight indent or control reveal — makes data surfaces feel alive at near-zero cost.
- **Live numbers:** counters ticking to real values on first view. Only real values — fake liveliness is a trust leak.
- **Navigation state change on scroll:** bar gains hairline/blur/compaction after the first scroll — communicates depth without stealing attention.
- **Drag physics:** dragged items lift (shadow+scale), targets highlight on approach, drops settle with ease-out.
- **Pull/overscroll responses (mobile):** rubber-banding and pull-to-refresh where the platform expects them — platform conventions are satisfaction patterns too.

6.4 When NOT to animate

- High-frequency micro-actions (typing echo beyond caret, per-cell table updates, rapid list scrolling).
- Anything on the critical path of user speed — power users must never wait for beauty.
- Layout thrash moments (opening 50 items at once) — batch to a single transition.
- When `prefers-reduced-motion` is set (Part 8.5) — full branch, not just "shorter".

PART 7 — FEEDBACK ON INTERACTIONS (the acknowledgment contract)

7.1 The contract

No user action may fall into silence. Every interactive element owes a complete state ladder:

default → hover (pointer devices) → focus-visible → active/pressed → loading/working → success or error → disabled — and where relevant: selected, dragging, dirty. A button is not one design; it is seven-plus designs. Shipping fewer means users will discover the missing one, and each discovery reads as cheapness.

7.2 Feedback timing (binds to Part 5.1)

- **<100ms: acknowledge.** Press states instant and local, CSS-only, never waiting for JavaScript or network.
- **<1s: just deliver the result.** Adding a spinner to sub-second operations makes products feel slower and jitterier — the flash-of-spinner anti-pattern.
- **1–10s: indicate at the point of action.** Button label swaps to gerund ("Saving…") with inline indicator; trigger disabled against double-submit; the rest of the page stays alive. Never a full-screen overlay for a local operation.
- **>10s or known-size: determinate progress** + current step name + cancel + (if possible) leave-and-notify.

7.3 Optimistic UI

For high-success-rate, low-stakes operations (toggle, like, rename, reorder, mark-done): apply the change instantly, sync in the background, and roll back with an apologetic, undo-bearing toast on rare failure. Perceived latency drops to zero. Never optimistic: payments, irreversible sends, deletions of substance, legal/financial actions.

7.4 Undo beats confirmation [STRONG — NN/g heuristic #3]

Routine confirmation dialogs train blind clicking (banner blindness for modals). Prefer immediate action + 5–10s undo affordance. Reserve confirmation for truly irreversible/high-stakes actions, and there make it *meaningful* friction: type-the-name, hold-to-confirm, or a deliberate ceremony — friction proportional to consequence, and friction that explains itself.

7.5 Channels

- **Visual:** first at the point of action, then (if needed) global: toast, badge, banner. Global-only feedback for local actions disorients.
- **Haptic (mobile apps):** light impact for confirmations, distinct pattern for errors; echo only real events. Gratuitous vibration destroys the channel's meaning.
- **Audio:** opt-in or absent in most products; if used, map to events 1:1 and respect silent mode.

- **Toasts:** one at a time; 4–6s; dismissible; carry the undo when undo exists; never the sole carrier of errors that require action (those go inline and persistent).
- **Badges/dots:** quantified where possible (3, not •); cleared by viewing, not by time.

7.6 Focus and keyboard feedback [LAW — WCAG 2.4.7, 2.4.11]

- `:focus-visible` ring on every focusable element, ≥2px, contrasting with both the component and its background, never clipped by `overflow:hidden` or hidden under sticky bars.
- Focus order = reading order; keyboard reachability = mouse reachability. One full Tab-only walkthrough per release is the cheapest audit in software.

7.7 System status beyond actions [NN/g heuristic #1]

Background processes announce themselves: syncing indicators, offline banners, autosave timestamps ("Saved just now"), session-expiry warnings *before* expiry, long-job completion notifications. Silence about status is indistinguishable from failure — users assume the worst.

PART 8 — SMOOTHNESS (motion engineering)

8.1 The 60fps contract

- Animate **only** `transform` **and** `opacity` — compositor-thread properties. Animating width/height/top/left/margin/padding/box-shadow/filter forces layout or paint and stutters on mid-range hardware, which is most hardware.
- Fake expensive effects: shadow "animation" = static shadow layer with animated opacity; height expansion = transform-based reveal or measured max-height with care; blur transitions = pre-blurred layer cross-fade.
- `will-change` is a scalpel: apply just before animation, remove after. Blanket `will-change` consumes memory and can *cause* jank.
- Test on throttled CPU (4–6×) — smoothness on a development machine is not evidence.

8.2 Duration DNA (define once per project, then never improvise)

- Micro feedback (press, toggle, checkbox): **80–150ms**
- Standard transitions (hover, dropdown, reveal, tooltip): **200–350ms**
- Structural moves (panel, drawer, modal, page transition): **350–600ms**
- Ceremonies (rare, weighty): up to **~1000ms**, earned by rarity.

Exits ~20–30% faster than entrances (leaving things get out of the way). Larger distances/elements get the longer end of each band.

8.3 Easing DNA (pick 2–3 named curves; forbid the rest)

- A signature ease-out for arrivals — e.g. `cubic-bezier(0.22, 1, 0.36, 1)` (fast start, long settle; reads refined) or Material's emphasized curves.
- A symmetric ease-in-out for reversible/spatial moves — e.g. `cubic-bezier(0.65, 0, 0.35, 1)`.
- Optionally one mechanical/instant curve for impacts.

- **Never raw** `linear` except continuous loops (spinners, marquees); **never default** `ease` for anything with intent. Mixed improvised easings across one product is an incoherent-feedback failure (Part 0.3 #4).

8.4 Choreography

- **Stagger** grouped entrances 40–120ms apart, cap ~5–6 items; beyond that, animate the group as one. Full-page per-element staggering is a template tell.
- **Sequence, don't pile:** container first, then contents; one primary motion at a time; simultaneous unrelated animations read as chaos.
- **Scroll-triggered > scroll-scrubbed** for reliability: IntersectionObserver reveals (threshold ~0.1–0.2, translate ≤30px, opacity 0→1, once). Scrubbed/parallax effects are a stylistic choice — execute with restraint (small deltas), test on touch, and never attach content-critical information to them.
- Respect scroll: no hijacking, no smooth-scroll override of user wheel, `scroll-margin` on anchor targets under sticky headers.

8.5 Reduced motion is a health requirement [LAW — WCAG 2.3.3]

```
@media (prefers-reduced-motion: reduce) {
  * { animation: none !important; transition: none !important; }
  [data-reveal] { opacity: 1; transform: none; }
}
```

Plus JS branches: skip observers and parallax, jump to final states, keep functional indicators (progress) but stop decorative ones. Vestibular disorders are common; this is not a preference toggle.

8.6 Perceived smoothness beyond animation

- **CLS ≈ 0:** reserve space for images (explicit dimensions), ads/embeds, and fonts (metric-compatible fallbacks). Layout shift reads as *broken*, which is worse than slow.
- Instant local acknowledgment (Part 7.2) buys more perceived speed than most real optimizations.
- Content-visibility/virtualization for long lists; paginate or virtualize before the browser janks, not after.

PART 9 — THE STATE INVENTORY (skeleton, empty, error, loading, offline — the real product)

9.1 The 12 states every dynamic screen owes you

A screen is not designed until every applicable state is designed. The ideal state is the *least seen* state in production; products feel premium in states 2–10:

1. **Ideal** — the mockup, with realistic (not flattering) content.
2. **Empty** — first use, nothing yet; also post-cleanup empty and filtered-to-zero (three different empties!).
3. **Loading** — initial, refresh, and pagination loading are distinct.
4. **Partial** — 1 item where 50 were designed for; short content in tall layouts.
5. **Error** — field-level, operation-level, and systemic (three tiers).
6. **Offline** — detection, banner, queued actions, recovery.
7. **Stale / refreshing** — showing old data while fetching new (never blank-then-reload).

8. **Success / confirmation** — the peak-end investment (Part 5.9).
9. **Disabled / locked** — permissions, plan limits, prerequisites; always with a *why* and a path forward.
10. **Overflow** — 10,000 rows, 300-character names, 15-deep nesting: truncation and pagination rules decided, not discovered.
11. **Interacted** — hover/focus/active/selected/dragging (Part 7.1).
12. **Dirty / unsaved** — pending edits, navigation guards, autosave indicators.

9.2 Loading: choosing the right indicator [STRONG — NN/g]

- **<1s:** show nothing. A spinner that flashes for 200ms adds jitter, not information.
- **1–10s, structural content loading: skeleton screens** — neutral shapes mirroring the true layout.
- Skeleton geometry must match the real content (matching rows, avatar circles, text bars of varying width). A skeleton that doesn't match the loaded layout is a bait-and-switch and *adds* perceived jank.
- Animate subtly (shimmer or pulse) — motion signals progress; static grey reads as frozen.
- Skeleton the *content*, never the chrome (nav/frame render instantly and real).
- Cap it: if load may exceed ~10s, switch to progress + messaging rather than eternal skeleton.
- **1–10s, discrete action:** inline indeterminate spinner at the point of action (in the button, next to the row) — not a page overlay.
- **>10s or known size (upload, export, batch): determinate progress bar** + step labels ("Uploading 3 of 8…") + cancel + leave-and-notify where possible. Progress never moves backwards; avoid parking at 99% — use honest step names for the long tail.
- Legitimate perception tools: begin bars at ~5–10% (0% looks dead), accelerate perceived progress near completion, rotate informative step text. Illegitimate: fake progress loops, invented counters — trust leaks.

9.3 Empty states are onboarding, not voids

Formula: **(1) what this space is → (2) what it looks like alive (sample, illustration, template) → (3) the one action that fills it.** "No items found" alone is a dead end and an execution failure. The three empties differ: first-use empty teaches; filtered-zero offers to clear filters; post-deletion empty confirms and points forward. Empty states are also brand moments — voice lives here.

9.4 Error states (the anatomy of a humane error)

1. **What happened**, in human words — "We couldn't save your changes", never code-first ("Error 0x80070057").
2. **Why**, if known — "You're offline", "That name is already taken".
3. **What to do next**, with the action attached — Retry button, field to fix (focused), status link, support contact with context pre-filled.
4. **Preserve all user input.** Destroying typed content is the cardinal sin of error design; forms survive refresh, drafts persist.
5. **Tone:** never blame ("Invalid input" → "That email needs an @"), never joke about lost work, never ALL-CAPS panic.

Placement tiers: field errors inline at the field + focus moved to first error; operation errors at the point of action; systemic errors as page-level banner. Keep technical detail behind a copyable "details" disclosure for support. Log codes silently.

9.5 Offline & resilience

Detect and announce (banner), queue what is queueable, disable what is not (with explanation), retry with exponential backoff, and *announce recovery* ("Back online — 3 changes synced"). Timeouts must exist: fail honestly by 10–30s. An infinite spinner is the single worst state in software — it converts patience into distrust at a fixed exchange rate.

9.6 Overflow & truncation (the forgotten state)

Per text element, decide: wrap, end-truncate + tooltip/full-view, middle-truncate (filenames, hashes — keep both ends), or line-clamp with expansion. Per collection, decide behavior at 10× expected volume: pagination, virtualization, or progressive "show more". Test with a 300-character name, an emoji-heavy title, a right-to-left string, and 10,000 rows before calling a screen complete.

PART 10 — MOBILE-FRIENDLY (thumbs, glare, interruptions)

10.1 The thumb zone [STRONG — Hoober's grip research]

Roughly half of phone use is one-handed thumb operation. The screen divides into: **easy** (bottom-center arc) — primary actions, main navigation, sticky CTAs; **stretch** (top-center, far edges) — secondary controls; **hard** (top corners) — nothing frequent, and *deliberately* the destructive (hard-to-reach = accidental-tap insurance). Bottom tab bars and bottom sheets exist because of this; frequent actions pinned top-left on a 6.7″ phone tax every single use.

10.2 Touch targets [LAW + platform guidance]

- **WCAG 2.2 (2.5.8) legal floor: 24×24 CSS px. Apple HIG: 44×44pt. Material: 48×48dp.** Ship 44–48; reserve 24 for genuinely dense pro UIs.
- The visual glyph may be smaller than the *hit area* — extend with padding or pseudo-elements, above all for icon buttons and close (×) targets.
- **≥8px between adjacent targets** — fat-finger errors come from proximity more than size.

10.3 Mobile input

- Correct keyboards via semantic types and `inputmode`: `type=email`, `type=tel`, `inputmode=numeric` for OTP/PIN/cards **[STRONG — Baymard]**; `autocomplete` attributes (`name`, `email`, `one-time-code`, `cc-number`…) — autofill is a conversion feature.
- **Inputs ≥16px font-size** or iOS zooms on focus and stays zoomed — the most common silent mobile-form defect.
- Labels **above** fields (thumbs and keyboards occlude side labels); never placeholder-as-label (vanishes on entry — recall failure, accessibility failure, review failure).
- Prefer native pickers, segmented choices, and steppers over dropdowns; each dropdown on mobile is a modal excursion.
- Autofocus with care: it summons the keyboard over half the viewport.

10.4 Layout & viewport

- Canonical design width **~390px**; verify at 360 (Android floor) and at large phones.

- `viewport-fit=cover` with `env(safe-area-inset-*)` padding — sticky bottom elements add `padding-bottom: env(safe-area-inset-bottom)` or the home indicator overlaps them.
- One column; tables become cards or intentional horizontal-scroll regions with a visible affordance (edge fade / peeking column).
- Stacking order on mobile is an *editorial* decision (content priority), not the accident of desktop DOM order.
- Sticky budget: one top bar + at most one bottom bar; stacked stickies can eat 40% of a small screen.
- **Hover does not exist.** Every hover-revealed control or tooltip needs a tap-visible equivalent — branch with `@media (hover: none)` / `(pointer: coarse)`. Hover-only information is amputated information for half your users.

10.5 Context & performance

- Test on throttled mid-range hardware and 3G-class networks — that is the median reality, not your flagship on fiber.
- Design for sunlight (contrast margins), interruptions (autosave, restore position and drafts on return), and short shallow sessions (front-load value; defer registration walls until value is demonstrated).
- Gestures need visible alternatives — gesture discoverability is near zero **[STRONG]**; never override system edge-swipes.
- Tap feedback replaces hover feedback: `:active` states become the primary satisfaction channel; keep them instant (Part 7.2).

PART 11 — DESKTOP-FRIENDLY (precision, keyboard, density)

11.1 What desktop uniquely offers

Precision pointer, physical keyboard, large canvas, long sessions, multi-window workflows. "Desktop-friendly" means exploiting all of these — not stretching the phone layout to 1440px and calling the leftover margins a design.

11.2 Hover as an information layer

- Hover states on **100% of interactive elements** — hover is how pointer users sonar-scan what is clickable.
- Reveal-on-hover secondary actions (row buttons, drag handles) keep dense surfaces calm — but every one must also appear on keyboard focus and have a mobile equivalent.
- Tooltips: ~300–500ms entry delay, instant exit, positioned to never cover the target's siblings; never the sole carrier of critical information.
- Cursor semantics: `pointer` for actions, `grab/grabbing` for draggables, `text` where selectable, resize cursors at real resize edges. Custom cursors: a signature-budget item, one per product, never on text.

11.3 Keyboard as a first-class citizen [LAW — WCAG 2.1.1]

- Everything mouse-doable is keyboard-doable: Tab order = visual order, Enter/Escape contracts on every dialog, arrow keys in menus and lists.
- Power accelerators for frequent actions: command palette (⌘K — now a Jakob's-Law convention in productivity tools), `/` for search, `?` for the shortcut overlay, Esc always retreats.

- Visible focus always (Part 7.6); focus returns to the trigger when dialogs close.

11.4 Using the width

- Canonical check at **1440**; verify 1280 and 1920+. Text never full-bleed (Part 2.2 line length).
- Spend width on: master-detail side-by-side, persistent navigation, comparison views, density options (comfortable/compact) — not on stretching cards into 600px-wide slabs.
- **Half-screen tiling is a desktop state:** your app at ~900px in a side-by-side window is common desktop reality — it must be a designed width, not an accidental tablet fallback.
- Long-session ergonomics: sticky headers for orientation, `scroll-margin` on anchors, remembered scroll positions, no scroll hijacking ever.

11.5 Desktop trust markers

Text is selectable; links are real links (middle-click/new-tab works); right-click is not hijacked without a superior replacement; drag-and-drop exists where objects imply it; undo history is deep; the browser back button always works and never loses state. These invisible conventions are what users mean when something "feels solid".

PART 12 — FORMS & INPUT (where trust and money are won or lost) [STRONG — Baymard]

1. **Every field must justify its existence.** Each field measurably costs completion. Optional fields: cut, or collapse behind an explicit "Add … (optional)" disclosure. Ask for information at the moment it is needed, not upfront.
2. **Labels above fields, permanently visible.** Placeholder-as-label fails memory (what was this field?), accessibility (no accessible name when filled), and review-before-submit. Placeholders are for *format examples* only ("e.g. 4111 1111…"), and must not pass contrast as if they were values.
3. **Validate on blur, not on keystroke** — premature red while the user is still typing is hostile. Once a field is in an error state, *re-validate on keystroke* so redemption is instant.
4. **Long forms:** error summary at top (linked to fields) + inline messages at each field + focus moved to the first error on submit attempt.
5. **Field design:** single "Full name" beats First/Last for most products (and respects global name formats); format-as-they-type masks for cards/phones (grouping reduces errors and reads as premium); appropriate widths that signal expected content length (a 60px ZIP field, not full-width).
6. **Buttons state their action:** "Create account", "Pay $49", never "Submit". Avoid disabled-until-valid as the only signal — users cannot discover *why* it's disabled; prefer enabled + validation on attempt, or pair disabled with a visible reason.
7. **Never lose user data.** Dirty-form navigation guards, draft autosave, refresh survival, and back-button safety. Lost input is the least forgivable failure in interface design.
8. **Multi-step flows:** show position and progress (goal-gradient, Part 5.10), make steps revisitable, never re-ask answered questions **[LAW — WCAG 3.3.7 redundant entry]**.
9. **Sensitive moments (payment, identity):** state expectations near the risky field ("charged after trial ends", "we never post without asking"), show trust markers in the field's line of sight, and never surprise with costs at the last step — the #1 documented cause of checkout abandonment **[STRONG — Baymard]**.

PART 13 — ACCESSIBILITY AS A QUALITY FLOOR [LAW — WCAG 2.2 AA]

Accessibility is not a legal tax appended at the end; every item below also improves the product for tired, distracted, one-handed, sun-blinded, low-bandwidth users. Curb cuts help everyone.

- **Contrast:** 4.5:1 body text; 3:1 large text, UI components, and meaningful graphics (1.4.3, 1.4.11).
- **Target size:** ≥24px, 44–48 recommended (2.5.8; Part 10.2).
- **Keyboard:** full operability, no traps, visible focus, focus not obscured by sticky UI (2.1.1, 2.4.7, 2.4.11).
- **Motion:** reduced-motion honored (Part 8.5); nothing flashes more than 3×/second (2.3.1).
- **Structure:** one `h1` per view; heading levels never skip; landmarks (`nav`, `main`, `footer`); alt text describes *function*, decorative images get `alt=""`.
- **Forms:** programmatic labels (`<label for>`), errors announced (`aria-live`), instructions not color-only or placement-only.
- **Color never the sole carrier of meaning** (1.4.1): status = word/icon + color, always.
- **Zoom:** the layout survives 200% browser zoom and 320px-equivalent reflow (1.4.10).
- Screen-reader sanity pass: can a user hear the page in order and operate every control? Semantic HTML gets you 90% of the way; ARIA is a repair kit, not a foundation.

PART 14 — CONTENT, MICROCOPY & TRUST DETAILS

14.1 Copy rules (copy IS interface)

- **Specifics over adjectives:** "Restores in under 2 seconds" beats "blazing fast"; "14-day trial, no card" beats "risk-free". Specificity is credibility; adjectives are slop's native language.
- Verbs over nouns; one idea per sentence; front-load the point. Headings never merely repeat what the section visually shows.
- Voice is a system like color: define 3–5 adjectives for the product's voice (e.g. calm, precise, warm) and audit copy against them, in errors and empty states especially — voice under stress is where brands are actually judged.
- **Microcopy at anxiety points:** beside destructive buttons, before payment fields, at permission prompts, at data-sharing moments. One calm sentence at the right coordinate outperforms a help center.

14.2 Data display craft

- Tabular numerals in tables/counters; consistent decimal precision; units always; midnight-crossing and timezone honesty in timestamps ("2h ago" with exact time on hover/tap).
- Identifiers middle-truncated (both ends visible); copy-to-clipboard affordances on anything users will need elsewhere, with confirmation feedback.
- Numbers users must compare sit in columns, right-aligned; numbers users must read sit in sentences.

14.3 The edges where users silently audit care

Favicon, social share card, page `<title>` with meaningful verbs, 404/500 pages, email templates, print styles, browser-tab behavior during loading, the selection color, the scrollbar in themed UIs. None are seen daily; all are seen eventually, and each broadcasts whether anyone decided anything.

14.4 Honest UI (dark patterns are compounding debt)

No fake urgency or invented scarcity; no confirm-shaming ("No thanks, I hate saving money"); no pre-checked upsells; no disguised ads; no roach-motel flows (easy in, deliberately hard out — cancellation must be as discoverable as signup). Beyond ethics: users recognize these patterns now, and each recognition permanently reprices the brand as cheap.

PART 15 — PERFORMANCE AS A DESIGN MATERIAL

- **Layout stability (CLS ≈ 0):** explicit dimensions on all media; reserved space for async content (that's what skeletons also do); metric-compatible font fallbacks with `font-display: swap`; no content jumping after first paint. Shift reads as broken; broken reads as cheap.
- **First impression speed (LCP):** the hero — whether type or image — renders fast: preload the display font and hero image, compress aggressively (modern formats), system-font-first stacks are a legitimate premium choice.
- **Interaction latency (INP):** main thread free to acknowledge input within 100ms — defer non-critical scripts, virtualize long lists, debounce expensive reactions but never the acknowledgment itself.
- Weight budgets set upfront (e.g. hero media ≤ X KB, total JS ≤ Y KB) — budgets are design constraints as real as brand colors, and they are exactly what makes products feel light in the hand.
- Every loading strategy in Part 9.2 is a *perceived*-performance tool; use them in concert with real optimization, not instead of it.

PART 16 — THE VERIFICATION RITUAL (how work is judged before it ships)

16.1 The hostile-critic pass (two rounds minimum, mandatory)

Adopt the voice of the meanest *competent* reviewer and attack the build:

- **Round 1 — structure:** Is there one dominant element per screen? Does hierarchy survive the squint test? Does any section exist only because a template had one? Would anyone remember anything tomorrow? Which of the five execution failures (Part 0.3) are present?
- **Round 2 — craft:** alignment drift, spacing-scale violations, contrast failures, orphaned words, easing/duration inconsistencies, missing states (run the Part 9 inventory element by element), copy adjectives without numbers.

Fix everything found. **Then run both rounds again** — the second full pass after fixes is where quality actually happens; a single pass always ships its own blind spots.

16.2 The three-width screenshot discipline

Verify every screen at **~390 / ~768 / ~1440** minimum (add 360 and ~900-tiled for coverage), as full-page captures inspected at **100% zoom** — downscaled previews hide spacing drift and rendering defects. Inspect element-by-element, not vibes-by-vibes: navigation, each section, footer, and every interactive state manually triggered (hover, focus, active, error, loading).

16.3 The pre-ship checklist (condensed, universal)

□ Foundational tokens decided and recorded (type/color/spacing/radius/shadow/motion) · □ one-sentence brief exists and screens audit against it · □ one dominant element per screen · □ full state inventory per dynamic surface · □ feedback ladder on 100% of interactive elements · □ motion uses only the named curves/durations · □ keyboard-only walkthrough clean · □ reduced-motion branch verified · □ contrast audit on real combinations · □ 44–48px touch targets, ≥8px gaps · □ 16px+ inputs, correct keyboards · □ 390/768/1440 screenshots clean at 100% · □ CLS ≈ 0, fonts swap gracefully · □ 300-char names, empty account, 10k rows survive · □ copy read aloud once (the ear catches what the eye forgives) · □ hostile critic ran twice.

PART 17 — DIRECTING AI (how to feed this codex to a coding agent)

1. **Feed the whole codex as system-level law**, then add the project's own taste layer separately: brand fiction, palette, typography, mood references. Codex = how to execute; taste layer = what world to build. Never let the agent infer either.
2. **Demand the foundations first:** require the agent to output its token decisions (type scale, palette roles, spacing scale, motion DNA) *before* any screen code, and approve them explicitly. This single gate eliminates most slop at the source.
3. **Require the state inventory per screen:** "list the states of this screen and show each" — agents default to ideal-state-only unless the inventory is demanded.
4. **Ban silent defaults:** instruct that any framework default kept must be declared as a *chosen* default with a one-line reason. Undeclared defaults are rejected in review.
5. **Review as a hostile tech lead:** approve with corrections, never rubber-stamp; name violations by codex part number ("Part 7.1: button missing focus state") — numbered laws end taste arguments.
6. **Iterate in concepts, then commit:** for identity-level work, request multiple distinct directions cheaply, choose one, *then* invest polish. Polishing an unchosen direction is waste; letting the agent self-select the direction is how template aesthetics sneak back in.
7. **Verification is part of the definition of done:** the agent runs Part 16 (three widths, two hostile passes, checklist) and reports findings *before* presenting work. "It runs" is not "it's done."

APPENDIX A — THE NUMBERS CARD (quick reference)

Topic	Value
Instant-feedback budget	100 ms
Flow limit / spinner threshold	1 s
Attention limit / progress-bar threshold	10 s
First impression formation	~50 ms
Interactive round-trip target	<400 ms
Body line length	45–75 ch
Body & input text minimum (mobile)	16 px
Contrast: body / large text & UI	4.5:1 / 3:1

Touch target: recommended / WCAG floor	44–48 px / 24 px
Gap between touch targets	≥ 8 px
Micro / standard / structural / ceremony motion	80–150 / 200–350 / 350–600 / ≤1000 ms
Exit vs entrance speed	exits ~20–30% faster
Stagger step / max staggered items	40–120 ms / ~5–6
Scroll-reveal translate cap	≤ 30 px
Verification widths	390 / 768 / 1440 (+360, ~900)
Toast duration / undo window	4–6 s / 5–10 s
Timeout ceiling before honest failure	10–30 s
Zoom survival	200% / 320px reflow

APPENDIX B — THE SLOP AUDIT (30 execution checks, any style)

Typography: □ unexamined default font □ timid scale □ full-width paragraphs □ body line-height on display type □ no tabular numerals in data. Color: □ role-less accent everywhere □ contrast failures on real combinations □ semantic color reused decoratively □ mixed neutral temperatures. Layout: □ arbitrary spacing values □ uniform section weighting □ alignment drift □ mixed radii/shadows without a scale. Interaction: □ elements missing hover/focus/active □ linear or default `ease` easing □ mixed motion DNA □ blocking animations □ hover-only affordances on touch. States: □ ideal-state-only screens □ flash-of-spinner □ skeleton mismatching real layout □ dead-end empty states □ code-first errors □ input lost on error □ infinite spinners. Copy: □ adjective claims without numbers □ headings repeating sections □ "Submit" buttons □ blame-tone errors. Trust: □ layout shift after paint □ fake urgency/dark patterns.

Scoring: each unchecked box is a defect; a premium product ships with zero. This audit is taste-blind — it never asks *what* style was chosen, only whether anyone was flying the plane.

End of codex. It regulates execution, never taste: bring any aesthetic — the codex will only ask whether every pixel of it was decided.